

Project direction changes:

In our final implementation, we removed the standalone image entity and its associated join tables. This decision simplified our schema and eliminated the complexities of image storage and deduplication. In general our final application follows and has functionalities that stated in the original proposal

Usefulness evaluation:

All core features outlined in our original proposal have been successfully implemented. The Home page has clear instructions in each section and a top-bar navigation with links to Query, New Report, Login, and My Reports. Users can search wildlife reports through the Query section using multiple filters (scientific name, Report ID, User ID, age approximation, and date range), with results displayed below. Authenticated users submit new wildlife reports via the New Report section, entering scientific name, age, time, and optional comments and the reports will be stored in the database. The authentication system allows registration with email, password, and a chosen User ID. Upon successful login, users receive a personalized "Welcome, <UserID>" greeting on the homepage and can exclusively access their submission history in the dedicated My Reports section. The Map that stated in the original proposal is not implemented in the final project

Data schema/ source changes:

Originally, the schema had a separate kingdom table, but in the final implementation Merged Kingdom_Name directly into the Phylum table and eliminated the standalone kingdom table. The planned image table was dropped due to implementation complexity. For our database design, in taxonomy in wildlife we changed entities like OrderId in Family table, FamilyId in Genus table etc. to ON DELETE CASCADE since if the upper category does not exist the lower will not exist either. Our data sources are <https://www.catalogueoflife.org/data/download> and <https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html>. Our data mostly use the catalogueoflife.org and there is no additional data source apart from those

ER diagram and Table Implementation changes:

Originally, the schema had a separate kingdom table, but in the final implementation Merged Kingdom_Name directly into the Phylum table and eliminated the standalone kingdom table. The planned image table was dropped due to implementation complexity. On our ER diagram is that in taxonomy in wildlife we changed entities like OrderId in Family table, FamilyId in Genus table etc. to ON DELETE CASCADE since if the upper category does not exist the lower will not exist either.

Functionality Added or Removed:

The image table is removed. All of the functionalities that stated in the original proposal is achieved

Advanced database program:

One of the advanced database programs that is implemented in the application is trigger. The verified user could click the verify button and after that it begins to update the attribute verified BIT from 0 to 1. Also the transaction is achieved when the user tries to update the database with a new report, if any of the name, age, time etc, is empty, the whole update will not execute. The way that updating the database with reports that users input is also a store procedure that could be applied to any input data.

Technical Challenges:

1. Miss match data is pushed into the table and the data want to be received
2. The UserId must be verified before updating on the database, it must exist
3. When searching for the reports, the scientific name, time and the species must be valid and match data stored in the database

Future improvements:

To enhance user experience, we would implement image functionality, which allows users to attach photos to wildlife reports, creating more intuitive visual documentation of sightings. Also, by implementing an image-classification AI model, enabling users to discover species names and look up the information simply by submitting photos.

Final labor division:

Nathan Colunga:

Focused on front end development and completed the interaction with backend services. Executed verification of userIds, species, and other inputted data from users in new reports and queries to ensure smooth connection from front end to backend to minimize conflicting inserts.

Yunqi Han:

The frontend section for first demo;

Writing the final report;

Fixing the issues from previous stage

Ryan Choi:

Worked on database setup along with designating primary keys, data types, and constraints. Setup Location table backend and assisted in debugging.

Subsection Stage 2:

The changes made regarding the feedback 1-5 could be checked on revised relational diagram.pdf in doc folder

1. The revised ERD.pdf is an updated version of relational diagram and all the attributes have been added.

2. The cardinalities between Species and Genus, Genus and Family are set to 1...*

3. Cardinality of the User side and species side are updated to 1..*

4. The Email VARCHAR(255) UNIQUE NOT NULL added

5. Image table is discard due to the technical issue

6. Step of normalization: In our previous design, the phylum, class, phylum, family, genus and species are the attributes in the same taxonomy table, in order to abide 3NF rule, since those are non-key attributes and they cannot depend on other non-key attributes within the same table. So it breaks up and generates 5 extra tables.

Feedback:

1. ERD does not show any attributes. It should. Also, the relationship names better be verb-like.

2. The cardinalities between Species and Genus, Genus and Family etc. these should be exactly 1 to many, not 0 or 1 to many as shown.

3. The cardinality of the User side of the "User who made report" and that of the Species side of Taxonomy of the reported file should probably be exactly 1, not 0 to 1.

4. Users should contain email, not specified in the description.

5. Images should probably be stored as url and GeometricShape should probably be a polygon with some latitudes and longitudes. Storing these as blob may complicate operations and impact performance.

6. Steps of normalization are not shown at all.

Subsection Stage 3:

1. The kingdomID has been changed to KingdomName and set to NOT NULL. The changes made regarding feedback 1-5 could be checked on revised database design.pdf in the doc folder.
2. The PhylumId, ClassId, OrderId, FamilyId, GenusId that in their sub-category table are set to ON DELETE CASCADE
3. User email is set to UNIQUE NOT NULL
4. The ImageFile BLOB has been removed, since the image table is discard
5. The GeometryPolygon BLOB is still included in the design as it was made for future development. The goal was to use the longitude and latitude positions of the user to systematically determine the geographic location and associated city using this stored database and polygon system in order to let the user not be as involved in the report process making it easier for data entry.

For feedback 6-9, there is a separate detailed document and shows how we fix the issues we mentioned in feedback. Refer to Stage 3 feedback 6-9.pdf in the doc folder

6. The number of rows we get from each of 4 advanced queries are shown in the screenshot, which are 46, 10, 1210, 622
7. The second one is searching the number of each species in a class reported by a certain user that specified by the user_id, detailed information could be found in the document that mentioned above
8. Three of our advanced queries show more than 15 rows. Only one of them has only 10 rows, there are not 15 rows here as the user that was queried did not submit enough reports to go past 15 rows. Detailed information could be found in the document.
9. Since we are using XAMPP and the embedded database system does not support EXPLAIN ANALYZE, we simply use EXPLAIN and compare the time queries executed after applying each index.

Feedback:

1. What is the KingdomID? I don't see a Kingdom table. Should it be the Kingdom name? In that case, this should probably be NOT NULL.
2. Shouldn't class, order, family, ... entities be on delete cascade upon their FKs? They shouldn't probably exist without their upper level (e.g., can genus exist without a order?). Also, if it helps, probably you can merge the genus and species table since they combinedly identify any living

creature.

3. User email should be UNIQUE and NOT NULL
4. You don't probably need ImageFile BLOB when you have the URL
5. What is the GeometryPolygon BLOB in location? I think you just need some latitudes and longitudes.
6. You have mentioned the number of rows in each table but the screenshots do not verify that. You could run a count(*) query on each table to prove this.
7. The purpose of your second query is not clear, the description seems same as the first one.
8. None of your screenshots show 15 rows, nor do you explain why so.
9. Creating indexes one by one (e.g., dropping the others before creating the new one) and the EXPLAIN ANALYZE results for each, with appropriate logical reasoning are missing.